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Rakib Hossan

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Research Interests

LLM reasoning: verifying reasoning steps against evidence; revising conclusions when evidence changes

Reinforcement learning for LLMs and agents when reliable rewards are scarce (test-time RL)

Multimodal AI (speech and vision): the same questions when evidence is audio or images

Factuality and hallucination; multilingual and low-resource NLP

Education

Apr 2023–Jan 2027 **B.Sc. Engg. in Computer Science & Engineering**, *Bangladesh University of Business and Technology (BUBT)*, Dhaka, Bangladesh
CGPA – **4.00/4.00** (9th semester running)
Dean's Award for Academic Achievement – Spring 2025, Fall 2025
Thesis: Claim Revision Benchmark (in progress)

2012–2014 **Pre-Sea Training in Marine Engineering**, *United Marine Academy*, Dhaka, Bangladesh
CGPA – 3.66

2012 **Higher Secondary Certificate (Science)**, *Dhaka College*
GPA – 5.00/5.00

2010 **Secondary School Certificate (Science)**, *Dhanmondi Govt. Boys High School*
GPA – 5.00/5.00

Publications

BLP @ AACL 2025 [1] **Rakib Hossan**, and Shubhashis Roy Dipta. "PromptGuard at BLP-2025 Task 1: A Few-Shot Classification Framework Using Majority Voting and Keyword Similarity for Bengali Hate Speech Detection." Proceedings of the BLP Workshop @ AACL-IJCNLP 2025. [PDF] [Code]

Research Projects

Ongoing **Claim Revision Benchmark (Thesis)**
Lead author (undergraduate thesis). With Shubhashis Roy Dipta, Mir Sazzat Hossain, MD Fahim
- Building a benchmark on how claims in a paper depend on each other, and whether LLMs update them when the results they depend on change.

Jul 2026–Ongoing **Agentic Test-Time Reinforcement Learning**, *Penta AI team*
Co-author. With MD Fahim, Mir Sazzat Hossain, Intesar Tahmid, Galib Mahmud Jim, Md Mahir Jawad
- Test-time reinforcement learning (TTRL) for LLM agents, where the reward is a vote over the model's own answers, so agents keep improving on new tasks without labels.

Jul 2026–Ongoing **Bangla Dialectal Medical ASR Benchmark**, *Penta AI team*
Lead author. Mentored by Mir Sazzat Hossain and MD Fahim
- A benchmark for clinically significant errors in Bangla dialectal medical speech recognition, built from synthetic doctor–patient conversations generated with LLMs and text-to-speech.

Jul 2026–Ongoing **Medical Prescription Understanding**, *Penta AI team*
Core contributor. With MD Fahim, Mir Sazzat Hossain, Intesar Tahmid, Galib Mahmud Jim, Md Mahir Jawad
- Reading prescription images with multimodal models, so that drug names, dosages, and instructions can be extracted reliably.

Industrial Experience

Apr 2026–Present **AI Researcher & Engineer**, *Penta Global Ltd.*, Dhaka
- Research since Jul 2026: agentic test-time RL; lead a Bangla dialect medical ASR benchmark; medical prescription understanding.
- Designed LLM pipelines for claim extraction, evidence grounding, and dependency-graph validation over scientific documents.
- Applied RAG, **majority-vote consensus** across repeated generations, and **LLM-judge** quality scoring to reduce hallucination.

Aug 2026–Nov 2026 **AI Engineer**, *AVNBD*, Dhaka
- Built a multi-stage LLM pipeline to extract, structure, and translate a large-scale human-rights dataset from a scanned Bangla source.
- Cross-referenced extracted records against an independent third-party dataset to validate accuracy.

Jun 2025–Feb 2026 **Software Engineer**, *OpenCare*, Hybrid
- Institution management, Keycloak auth, MinIO file storage, Liquibase migrations, Dockerized deployment. [Contributors]

Apr 2024–Mar 2026 **Software Engineer**, *Penta Global Ltd.*, Dhaka
- Enterprise applications: Bangladesh Election Commission web system, ISP billing services, and the HRM Management System.

Honors & Awards

Jan 2026 **Dean's Award for Academic Achievement**, Faculty of Engineering & Applied Sciences, BUBT (Fall 2025)

Aug 2025 **Dean's Award for Academic Achievement**, Faculty of Engineering & Applied Sciences, BUBT (Spring 2025)

Skills

LLM Research LLMs, RL for LLMs (PPO, DPO, test-time RL), RAG, LLM agents, prompt engineering, few-shot learning, LLM-as-judge evaluation

ML & NLP Transformers, CNNs, RNNs/LSTMs, text classification, word embeddings

Frameworks PyTorch, Hugging Face Transformers, TensorFlow, Keras, scikit-learn, Pandas, NLTK

Languages Python, Java (Spring Boot), TypeScript (NestJS), C/C++

Systems PostgreSQL, Redis, RabbitMQ, Keycloak, Docker, CI/CD

Certifications

Apr 2026 Natural Language Processing Specialization, DeepLearning.AI

Mar 2026 Deep Learning Specialization, DeepLearning.AI

Feb 2026 Machine Learning Specialization, DeepLearning.AI

Prior Career: Marine Engineering

Five years at sea on super tankers before computer science: safety-critical, team-coordinated work where every fault is traced to a root cause.

Mar 2022–Jan 2023 **Junior Engineer**, *MT. XIDI, Seacon Star*, At sea
- Ran and maintained engine, power, boiler and water systems on a VLCC crude oil supertanker.

Sep 2017–Aug 2021 **Engine Cadet**, *ATLANTIC 1 (Jhampur Group), MT. ARTAVIL (NITC)*, At sea
- Four years on super tankers maintaining propulsion, power and emergency systems.

References

Available on request.